Software Requirements Specification (SRS) Document

Rural Agripreneur
Team 9
Naman Singhal, Samyak Jain, Samarth Singla, Shreyash Lohare, Ipsit Subuddhi

Brief problem statement

The current agricultural supply chain in Telangana is fragmented by multiple layers of intermediaries resulting in inflated costs for farmers and a lack of accountability regarding seed quality. Furthermore, farmers suffer from a knowledge gap lacking access to real-time, expert advice on soil suitability. This project addresses these issues by building a "Direct to Village" aggregator platform. It empowers Rural Agripreneur Executives (RAEs) to consolidate local demand and place bulk orders directly with manufacturers ensuring fair prices and quality while simultaneously providing a mentorship channel via District Advisors (SMEs).

System requirements

To ensure the system functions effectively in rural environments, the following requirements are defined:

Hardware Requirements:

- RAE Device: Low-cost Android Smartphone (Android 8.0+, 2GB RAM minimum).
- Admin/SME Station: Desktop or Laptop with a modern web browser (Chrome/Edge).
- Server: Cloud-hosted instance (AWS/GCP/Azure) capable of running Python containers and a PostgreSQL database.

Software & Technology Stack:

- Mobile App (RAE): React Native (Expo) for cross-platform capability; SQLite/Redux Persist for Offline-First local storage.
- Web Dashboard (Admin/SME): React.js with Tailwind CSS.
- Backend: Python (FastAPI) for high-performance API handling.
- Database: PostgreSQL (Relational Data) + Redis (Caching).

Network Constraints:

- The mobile application must support Offline Mode, allowing full order placement functionality without internet access, and auto-syncing data when a 3G/4G connection is restored.

Users profile

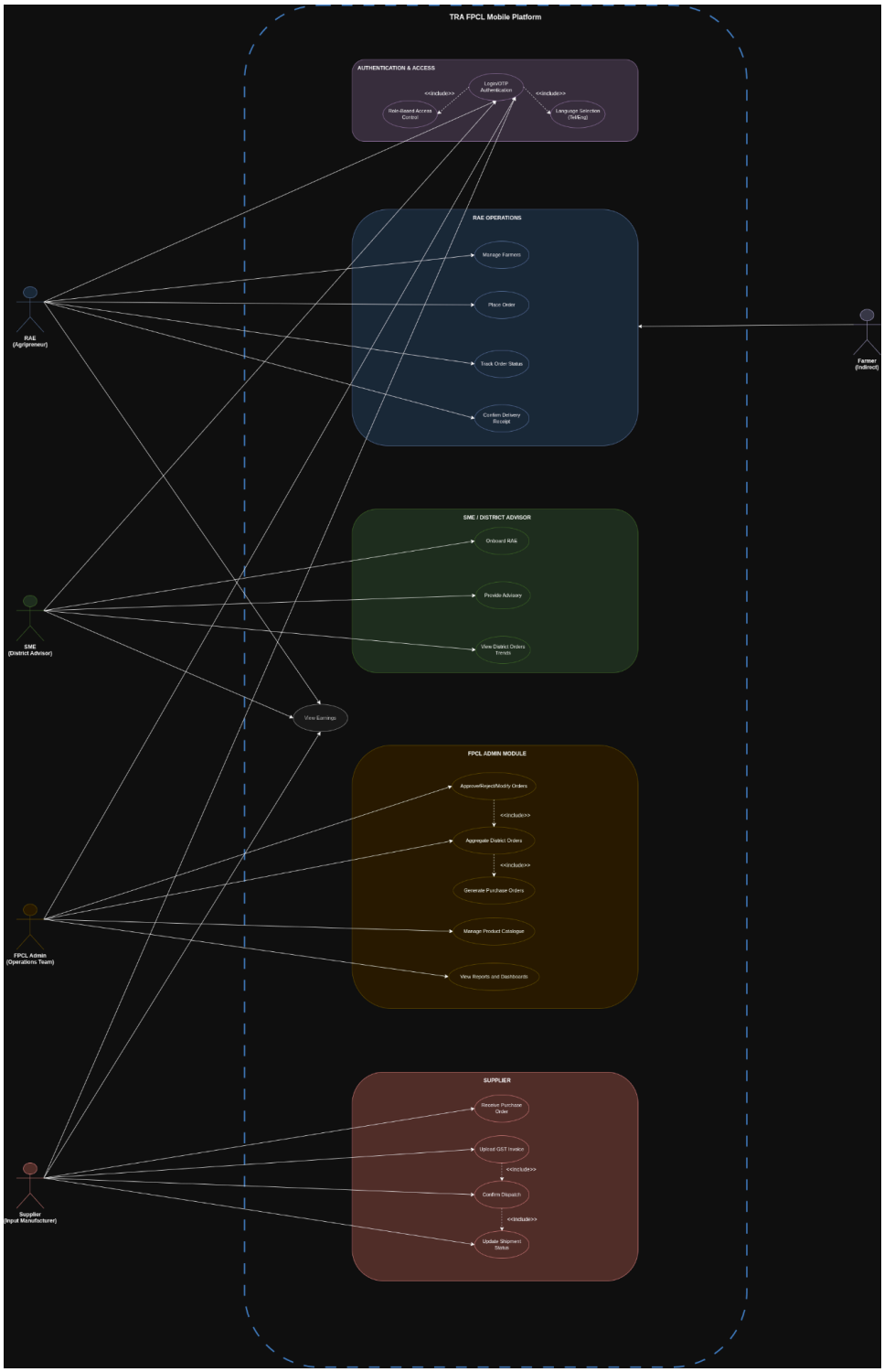
The system caters to four distinct user roles with varying levels of technical literacy:

1. Rural Agripreneur Executive (RAE):
    - Profile: Educated rural youth
    - Mode: Mobile App (Android)
    - Tech Literacy: Moderate tech literacy, native Telugu speakers
    - Constraints: Uses the app in fields with intermittent connectivity, requires a visual icon-driven interface with bilingual support (Telugu/English)
2. District Advisor (SME):
    - Profile: Experienced progressive farmers or domain experts
    - Mode: Mobile App for chat, mentorship and monitoring (Android)
    - Tech Literacy: Mixed
    - Role: Identifies and onboards RAEs, provides agronomic advice via chat
3. FPCL Admin:
    - Profile: Central operations team
    - Mode: Web Dashboard (Desktop)
    - Tech Literacy: High (comfortable with ERPs and Excel)
    - Role: Aggregates village orders into bulk Purchase Orders (POs), manages Master Data of products
4. Input Supplier:
    - Profile: External Manufacturer/Distributor
    - Mode: Mobile App (Android)
    - Literacy: Corporate/High
    - Role: Receives POs, uploads GST Invoices and updates Dispatch details

Feature requirements (described using use cases)

No.	User Case Name	Description	Release
1.	Login/ OTP Authentication	Users authenticate using mobile numbers and OTP. Role-based access determines module visibility.	R1
2.	Manage Farmers	RAE creates and maintains farmer records including land size, crop type, and village details.	R1
3.	Place Order	RAE selects products, associates them with a farmer and confirms the order. Data is stored locally if the internet is unavailable.	R1
4.	Track Order Status	RAE monitors order lifecycle from approval to delivery.	R1
5.	Confirm Delivery Receipt	RAE confirms that the farmer has received the delivered products.	R1
6.	Onboard RAE	SME registers a new RAE and assigns them to a village/mandal.	R1
7.	Approve/ Reject / Modify Orders	Validate incoming RAE orders.	R1
8.	Provide Advisory	SME provides technical agricultural guidance to RAEs.	R1
9.	Aggregate District Orders	Consolidate approved orders into district demand.	R1
10.	Generate Purchase Orders	Generate consolidated PO for supplier.	R1
11.	Receive Purchase Order	Supplier receives generated PO.	R1
12.	Confirm Dispatch	Supplier confirms shipment dispatch.	R1
13.	Update Shipment Status	Supplier updates shipment lifecycle.	R1
14.	View District Order Trends	SME monitors aggregated district-level demand and performance metrics.	R2
15.	View Earnings	RAEs and SMEs view a dashboard showing their accrued commissions based on delivered orders.	R2
16.	Generate GST Invoice	The system generates a PDF invoice for the RAE/Farmer including HSN codes and tax splits.	R2
17.	Manage Product Catalogue	Manage pricing, images, and product metadata.	R2
18.	View Reports & Dashboards	System-wide analytics and KPI monitoring.	R2
19.	Upload GST Invoice	Supplier uploads invoice documentation.	R2

Use case diagram



Link for clear image- https://drive.google.com/file/d/184sIJLxrUeLwbAo6c0utZbx7NRAfoqjU/view?usp=sharing

Use case description

UC-01 - Login / OTP Authentication

Use Case Number:	UC-01
Use Case Name:	Login / OTP Authentication
Overview:	Users authenticate using mobile number and OTP. Role-based access determines system permissions.
Actors:	All Users
Precondition:	User is registered in the system.
Flow (Main Success):	1. User enters mobile number. 2. System sends OTP. 3. User enters OTP. 4. System validates OTP. 5. System grants role-based access.
Post Condition:	User is authenticated and redirected to role-specific dashboard.

UC-02 - Manage Farmers

Use Case Number:	UC-02
Use Case Name:	Manage Farmers
Overview:	RAE creates and maintains farmer records.
Actors:	RAE
Precondition:	RAE is logged in.
Flow (Main Success):	1. RAE selects 'Manage Farmers'. 2. RAE enters farmer details. 3. System validates inputs. 4. System stores farmer record.
Post Condition:	Farmer record is stored in the database.

UC-03 - Place Order

Use Case Number:	UC-03
Use Case Name:	Place Order
Overview:	RAE creates an order associated with a farmer.
Actors:	RAE
Precondition:	RAE is logged in and farmer exists.
Flow (Main Success):	1. RAE selects products. 2. RAE associates products with farmer. 3. RAE confirms order. 4. System stores order.
Post Condition:	Order is created and stored.

UC-04 - Track Order Status

Use Case Number:	UC-04
Use Case Name:	Track Order Status
Overview:	RAE monitors the lifecycle of an order.
Actors:	RAE
Precondition:	Order exists in system.
Flow (Main Success):	1. RAE selects order. 2. System displays order status.
Post Condition:	RAE views current order status.

UC-05 - Confirm Delivery Receipt

Use Case Number:	UC-05
Use Case Name:	Confirm Delivery Receipt
Overview:	RAE confirms delivery of products to farmer.
Actors:	RAE
Precondition:	Order status is 'Delivered'.
Flow (Main Success):	1. RAE selects delivered order. 2. RAE confirms receipt. 3. System updates order status.
Post Condition:	Order marked as completed.

UC-06 - Onboard RAE

Use Case Number:	UC-06
Use Case Name:	Onboard RAE
Overview:	SME registers and assigns a new RAE.
Actors:	SME
Precondition:	SME is logged in.
Flow (Main Success):	1. SME enters RAE details. 2. SME assigns village/mandal. 3. System creates RAE account.
Post Condition:	New RAE account is active.

UC-07 - Approve / Reject / Modify Orders

Use Case Number:	UC-07
Use Case Name:	Approve / Reject / Modify Orders
Overview:	Admin validates RAE orders.
Actors:	Admin
Precondition:	Orders are submitted by RAEs.
Flow (Main Success):	1. Admin reviews order. 2. Admin approves/rejects/modifies. 3. System updates order status.
Post Condition:	Order status updated accordingly.

UC-08 - Provide Advisory

Use Case Number:	UC-15
Use Case Name:	Provide Advisory
Overview:	SME provides technical advisory support.
Actors:	SME
Precondition:	SME and RAE logged in.
Flow (Main Success):	1. RAE initiates query. 2. SME responds. 3. System stores chat history.
Post Condition:	Advisory session recorded.

UC-09 - Aggregate District Orders

Use Case Number:	UC-08
Use Case Name:	Aggregate District Orders
Overview:	Admin consolidates approved orders.
Actors:	Admin
Precondition:	Orders are approved.
Flow (Main Success):	1. Admin selects district. 2. System aggregates orders. 3. System prepares consolidated demand.
Post Condition:	District demand summary generated.

UC-10 - Generate Purchase Orders

Use Case Number:	UC-09
Use Case Name:	Generate Purchase Orders
Overview:	System generates purchase order for supplier.
Actors:	Admin
Precondition:	District orders aggregated.
Flow (Main Success):	1. Admin triggers PO generation. 2. System generates PO document. 3. Supplier notified.
Post Condition:	Purchase Order created and sent.

UC-11 - Receive Purchase Order

Use Case Number:	UC-10
Use Case Name:	Receive Purchase Order
Overview:	Supplier receives PO notification.
Actors:	Supplier
Precondition:	PO is generated.
Flow (Main Success):	1. Supplier logs in. 2. Supplier views PO details.
Post Condition:	Supplier acknowledges PO.

UC-12 - Confirm Dispatch

Use Case Number:	UC-11
Use Case Name:	Confirm Dispatch
Overview:	Supplier confirms shipment dispatch.
Actors:	Supplier
Precondition:	PO accepted.
Flow (Main Success):	1. Supplier enters dispatch details. 2. System records shipment.
Post Condition:	Shipment status updated.

UC-13 - Update Shipment Status

Use Case Number:	UC-12
Use Case Name:	Update Shipment Status
Overview:	Supplier updates shipment progress.
Actors:	Supplier
Precondition:	Shipment is in transit.
Flow (Main Success):	1. Supplier updates status. 2. System reflects update.
Post Condition:	Shipment lifecycle updated.

UC-14 - View District Order Trends

Use Case Number:	UC-16
Use Case Name:	View District Order Trends
Overview:	SME views analytics for district performance.
Actors:	SME
Precondition:	Order data exists.
Flow (Main Success):	1. SME selects analytics dashboard. 2. System displays trends.
Post Condition:	District performance displayed.

UC-15 - View Earnings

Use Case Number:	UC-17
Use Case Name:	View Earnings Dashboard
Overview:	Users view accrued commissions.
Actors:	RAE/SME
Precondition:	Orders completed.
Flow (Main Success):	1. User selects earnings dashboard. 2. System calculates commissions. 3. System displays earnings.
Post Condition:	Earnings data displayed.

UC-16 - Generate GST Invoice

Use Case Number:	UC-18
Use Case Name:	Generate GST Invoice
Overview:	System generates GST-compliant invoice.
Actors:	Admin/System
Precondition:	Order finalized.
Flow (Main Success):	1. Admin triggers invoice generation. 2. System generates invoice PDF.
Post Condition:	GST invoice created.

UC-17 - Manage Product Catalogue

Use Case Number:	UC-19
Use Case Name:	Manage Product Catalogue
Overview:	Admin manages product details and pricing.
Actors:	Admin
Precondition:	Admin logged in.
Flow (Main Success):	1. Admin adds/edits product. 2. System validates data. 3. System updates catalogue.
Post Condition:	Product catalogue updated.

UC-18 - View Reports & Dashboards

Use Case Number:	UC-20
Use Case Name:	View Reports & Dashboards
Overview:	Admin monitors system-wide KPIs.
Actors:	Admin
Precondition:	Data available in system.
Flow (Main Success):	1. Admin selects reports. 2. System displays KPIs.
Post Condition:	Reports displayed.

UC-19 - Upload GST Invoice

Use Case Number:	UC-21
Use Case Name:	Upload GST Invoice
Overview:	Supplier uploads GST invoice document.
Actors:	Supplier
Precondition:	PO processed.
Flow (Main Success):	1. Supplier uploads invoice. 2. System validates format. 3. System stores invoice.
Post Condition:	Invoice stored in system.